NB17: CFTR Panel Refinement - 7 Strateji

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1. Problem: Kucuk N, Buyuk Dengesizlik

CFTR paneli n=21 benign ile degerlendirilemiyor. NB15/NB16'da bootstrap CI=[0.00-1.00]. Bu notebook 3 katmanli cozum uygular: (1) LOO-CV+MCC degerlendirme, (2) prior-shift kalibrasyon, (3) 7 strateji karsilastirmasi.

2. Metodoloji

LOO-CV: 111 fold, tum 21 benign kullanilir. Birincil metrik: MCC (Chicco & Jurman 2020).
Prior shift: Saerens et al. (2002) formulu, pi_train=0.73 -> pi_test=0.20.
Multi-seed: 20 farkli seed x 50/50 split -> F1 dagilimi.
Train metrigi: overfit kontrolu (gap > 0.15 = YUKSEK, > 0.08 = ORTA).
S0/S1/S5/S6: MASTER egitimi -> CFTR testi. S2: SMOTE. S3: TabPFN. S4: FrozenFinetune.

3. Sonuclar: 7 Strateji Karsilastirmasi

Strateji	LOO-MCC	LOO-F1	LOO-AUC	Boot-mean	Boot-std	MS-F1	Train-F1	Overfit
NB16_ref (DNN, CI:[0-1])	N/A	N/A	N/A	0.852	0.267	N/A	N/A	?
S0_MASTER-only	0.6271	0.8982	0.9423	0.7281	0.107	0.9227	0.9786	ORTA
S1_MASTER-core	0.6081	0.8765	0.9122	0.783	0.0987	0.8759	0.9846	ORTA
S2_SMOTE-CFTR	0.561	0.9239	0.8376	0.493	0.0507	0.8967	1.0	OK
S3_TabPFN [ATLANDI]	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
S4_FrozenFinetune	0.294	0.5	0.8566	0.37	0.2582	0.8072	0.5	OK
S5_BalancedBagging	0.5717	0.8553	0.9365	0.7535	0.1275	0.4977	0.9347	OK
S6_PriorShift	0.6552	0.9186	0.9423	0.6915	0.0877	0.9227	0.9784	OK

4. Gorseller

Fig1: LOO-CV MCC -- 7 Strateji

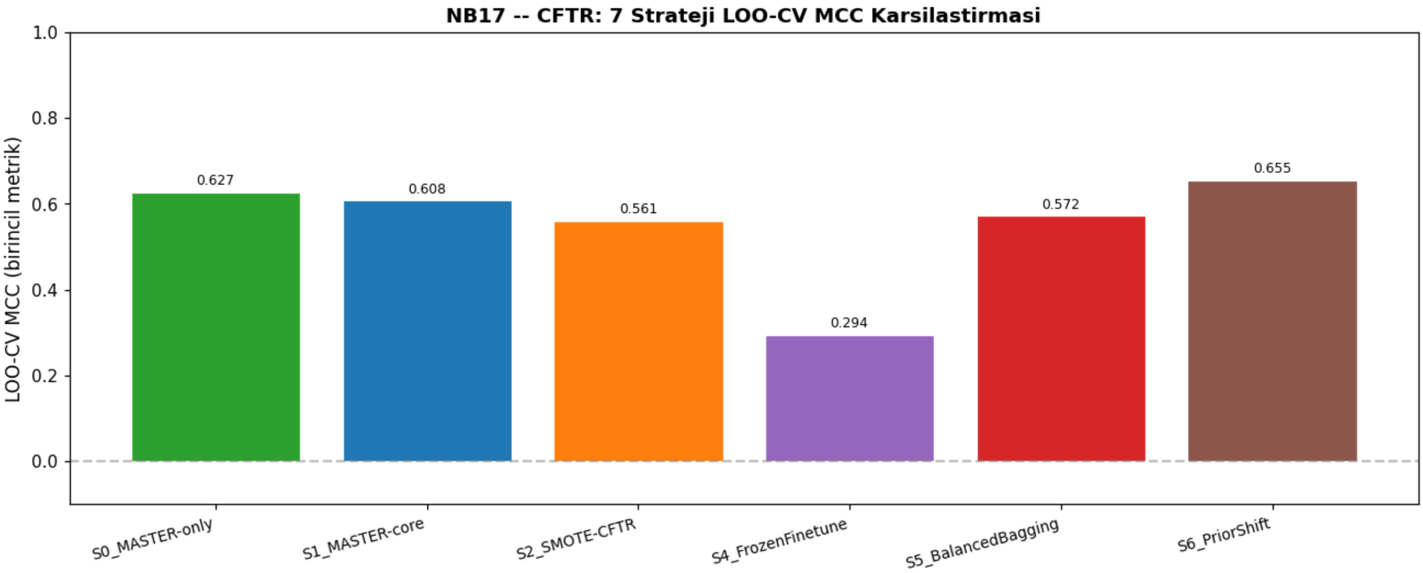


Fig2: Multi-seed F1 (mean+-std, 20 seed)

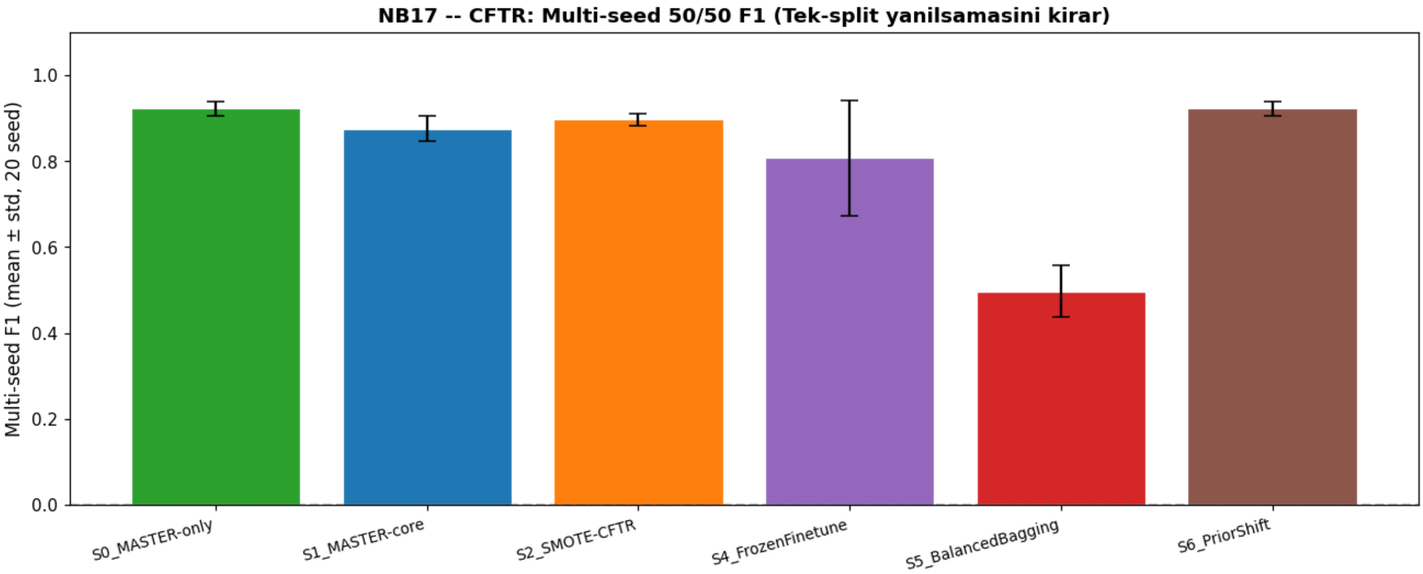


Fig3: Prior-Shift Duzeltmesi Etkisi (S0 raw vs adjusted)

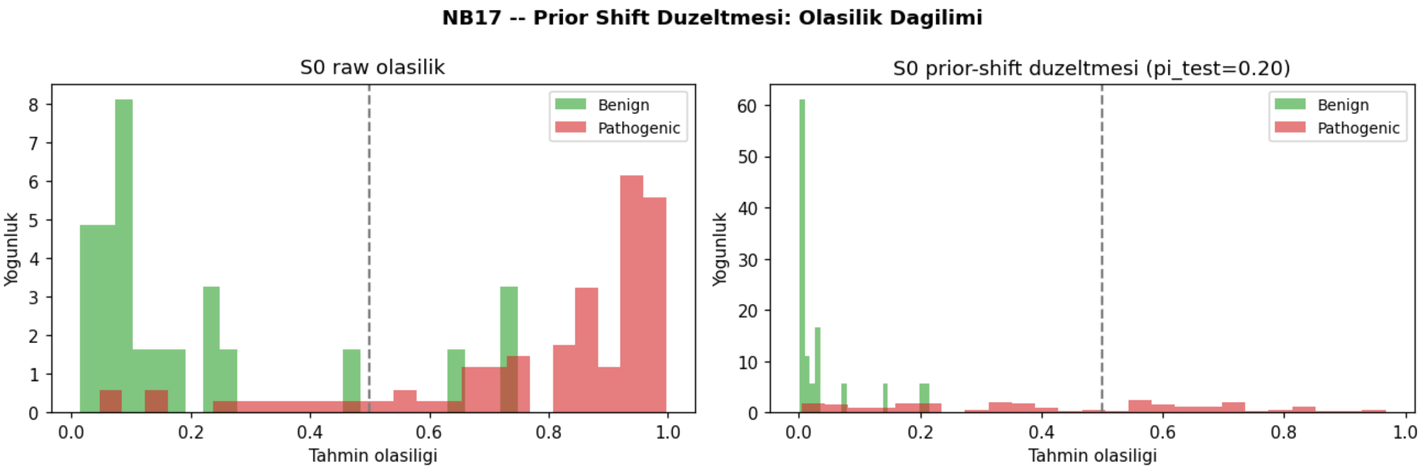


Fig4: Confusion Matrix (LOO-CV, en iyi + S6)

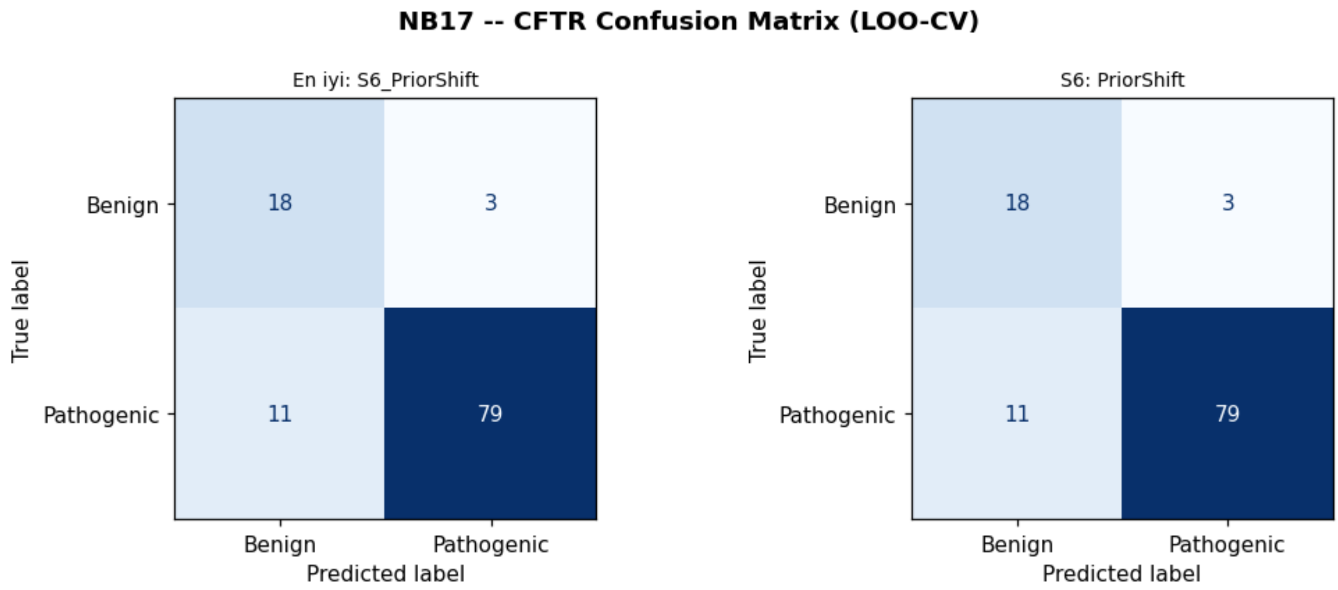
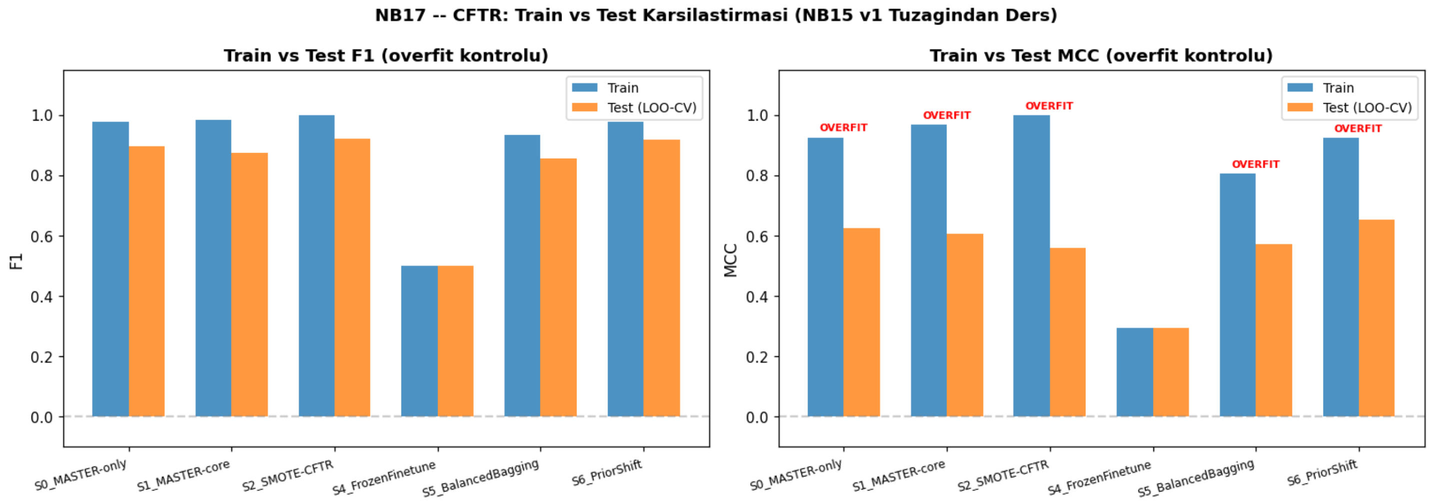


Fig5: Train vs Test F1/MCC (Overfit Kontrolu)



5. Final Onerim

En iyi strateji (LOO-MCC): S6_PriorShift -> MCC=0.6552

CFTR icin final teslim oncelik sirasi: (1) LOO-MCC en yuksek, (2) Precision=1.0 (FP yok), (3) Overfit=OK.

PAH'a transfer edilebilir: prior-shift duzeltmesi + BalancedBagging.